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# T-SQL Tips & Tricks



SQLRV, LLC

HEALTHCARE INFORMATICS

https://www.sqlrv.com



# Just who is this guy?

SQLRV, LLC
HEALTHCARE INFORMATICS

Principal Data Architect, Population Health

BYU PATHWAY WORLDWIDE

Adjunct Instructor

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Speaker – various events

B.S. Computer Science

M.S. Computer Information Systems

M.S. Health Informatics

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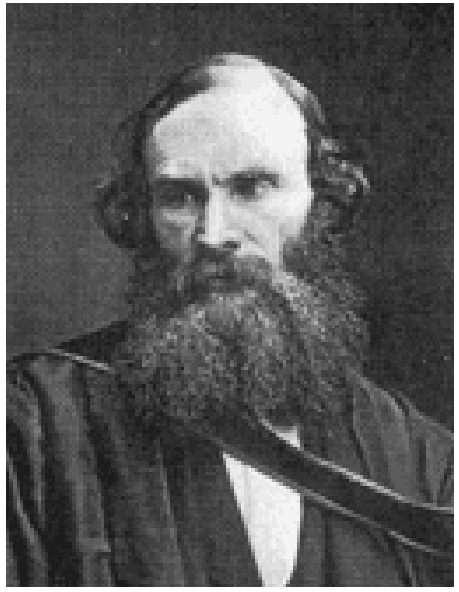
Doctor of Healthcare Administration

40 Years



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# Something to consider…



"If you cannot measure it, you can not improve it.“

-- Lord Kelvin (Sir William Thomson)

# What are Measures?

- Various types of measures
  - Clinical Quality Measures – Determine the quality of patient health care and adherence to quality standards
  - Utilization Measures – Determine the cost or performance efficiency of a health care facility
- Example Scenario
  - 2.5 million patients (85 million clinical events)
  - 5 Hospital Networks
  - 20 Clinical Quality Measures
  - Calculation time: 45 minutes



# What makes it possible?

- Database architecture
  - Smart table structure design (de-normalize where applicable)
  - Appropriate data types
  - Smart use of indexes and their management
  - Routine maintenance
- Efficient T-SQL code
  - Set-based operations
  - Consider performance on large data sets
  - Make use of various tips & tricks



# Mismatched data types

- Use only appropriate data types
  - Save space & time
  - Keep as simple as possible
- Avoid implicit data conversions
  - Automatic conversion from smaller to large data type
  - Performed on EVERY comparison, assignment, etc.
  - Can kill index operations resulting in full table scan
  - Can cause problems down the road with unexpected data



# Mismatched data types

- SBD – Silent But Deadly
  - Kills performance without warning
  - Hard to find killer – like a Ninja!
- Good reference links:
  - http://www.sqlconsulting.com/news1210.htm
  - http://www.sqlservercentral.com/articles/Editorial/72756/
  - http://www.sqlperformance.com/2013/04/t-sql-queries/implicit-conversion-costs
- Demo File #1



# User-Defined Scalar Functions

- Can turn set-based operation into a cursor
- Kills opportunity for parallelism
- Avoid use if at all possible
  - Incorporate into existing code
  - Use comments to identify code segment
  - Implement as Table-Valued Functions instead
- Good reference links:
  - http://sqlmag.com/database-performance-tuning/small-code-changes-big-sql-server-performance-improvements
  - http://www.sqlservercentral.com/articles/T-SQL/91724/
- Demo file #2



# Monitor performance

- Execution Plans, missing indexes, and unused indexes
- Log query activity with start/stop times and rows affected
- Compare performance with previous runs and trend over time
- Good reference links:
  - http://www.sqlservercentral.com/articles/Performance+Tuning/monitoringperformance/1007/
  - http://support.microsoft.com/kb/298475
- Demo file #3



# Common Table Expressions

- Simplify queries and increase readability
- Perform recursive functions (be careful here)
- Limit data to be processed
- Good reference links:
  - http://www.sqlservercentral.com/articles/T-SQL/2926/
  - http://technet.microsoft.com/en-us/library/ms186243%28v=sql.105%29.aspx
  - http://stackoverflow.com/questions/15301712/how-to-develop-a-recursive-cte-in-t-sql
- Demo file #4



# Common Table Expressions (cont.)

- Reduce data for testing purposes
- Eliminate duplicate data
- Protect sensitive data
- Test against random sample records
- Good reference links:
  - http://www.mindfiresolutions.com/How-to-delete-duplicate-rows-in-a-table-using-CTE-1461.php
  - http://www.sqlservercentral.com/blogs/sqlrnnr/2011/03/21/dedupe-data-cte/
- Demo file #5



# Tally Table

- Turn loops into set-based statements
- Avoid cursors
- Improve performance
- Good reference links:
  - http://www.sqlservercentral.com/articles/Tally+Table/70735/
  - http://www.sqlservercentral.com/articles/Tally+Table/70738/
  - http://www.sqlservercentral.com/articles/T-SQL/62867/
  - http://www.sqlservercentral.com/scripts/tally/153313/
- Demo file #6



# Conditional Joins

- Put conditions in table join – not WHERE clause
- Restrict data pulled by joined table
- Use one or more outer joins in OR situations (OR clause invalidates index usage)
- Good reference links:
  - http://weblogs.sqlteam.com/jeffs/archive/2007/04/03/Conditional-Joins.aspx
  - http://stackoverflow.com/questions/15660765/sql-server-conditional-join-based-on-column-values
  - http://www.sqlservercentral.com/articles/T-SQL/63322/
- Demo file #7



# T-SQL Tips & Tricks

- Situation: Need to reduce IO and improve performance on complex queries
- Problem: Query processing order of precedence
  - FROM (including joins)
  - WHERE
  - GROUP BY
  - HAVING
  - SELECT
    - Evaluate Expressions
    - Remove Duplicates (DISTINCT)
  - ORDER BY
  - OFFSET-FETCH/TOP
- Solution: Reduce data selected
  - Use proper criteria selection at appropriate points
  - Demo file #8



# T-SQL Tips & Tricks

- Situation: Merge incoming data to existing data
- Problem: Several factors to consider
  - Large data set with large record size
  - Most fields are text
  - Insert records if new
  - Update only when data changed
- Solution: Use data hash for comparisons
  - Quick & Easy indicator of changed data
  - Use staging table for easy operation
  - Demo file #9



# T-SQL Tips & Tricks

- Situation: Perform multiple operations on same set of filtered data
- Problem: Several factors to consider
  - Large data set with multiple related tables
  - Data identified by multiple value sets
  - Queries rely upon the order data appears
- Solution: Use single query to prepare final data
  - Make use of multiple CTEs
  - Use ROW_NUMBER() operation
  - Use appropriate JOIN criteria
  - Demo file #10



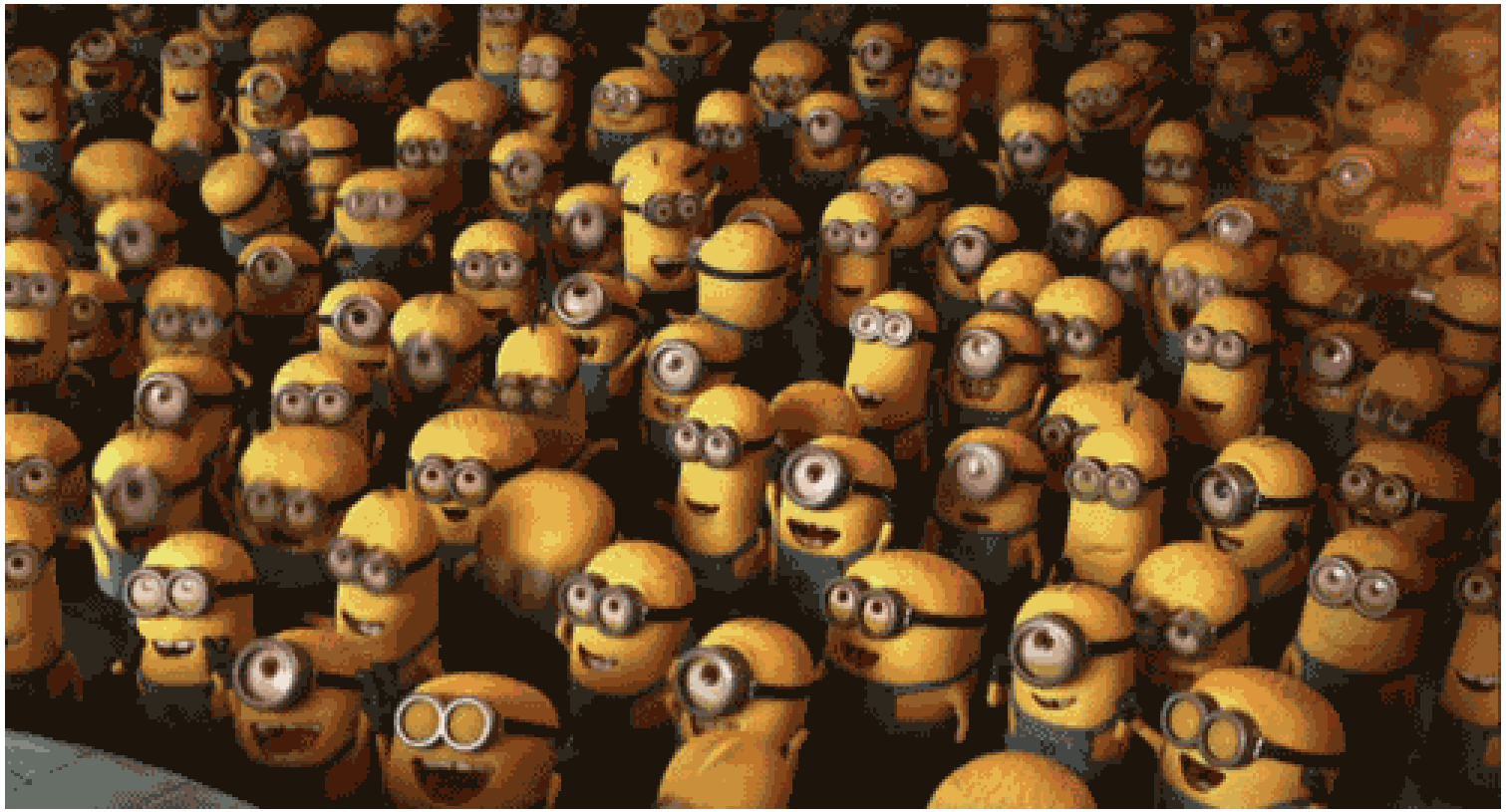
# Best Practices

- Programming Standards
  - Help to avoid religious wars in methodology
  - Increases code clarity and consistency
  - Enhances team interaction and efficiency
- Make things table driven as much as possible
  - Reduce “hard coding” wherever possible (document where not)
  - Simplifies code and enhances re-usability
  - Makes changes simpler and quicker to implement
- Code Management
  - Put common routines into stored procedures
  - Increase code reuse and reduce code maintenance
  - Remember: OOP isn’t just for .NET and Java



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